

MATH403: Homework 5

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October 2, 2024

6. Give an example of a group G and subgroups H and K such that $HK = \{hk \mid h \in H, k \in K\}$ is not a subgroup of G .

Proof. Consider $G = S_3$, the symmetric group of 3 elements. G contains the following elements

$$G = \{e, (12), (13), (23), (123), (132)\}$$

Now suppose we define the following subgroups

$$H = \{e, (12)\}$$

$$K = \{e, (13)\}$$

Observe that

$$HK = \{e, (12), (13), (12)(13)\} = \{e, (12), (13), (132)\}$$

However, $(13)(132) = (23)$ which isn't in HK , and so HK is not closed and so it is not a subgroup, even though H and K are. $\square$

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8. Let a and b be elements of a group G and H and K be subgroups of G . If $aH = bK$, prove that $H = K$.

Proof.

$\implies$ Let us show that $H \subseteq K$. Let $x \in H$. By **Property 1 of Lemma 7.1**, $ax \in aH$. Since $aH = bK \implies ax \in bK$. Furthermore, observe that

$$a \in aH \implies ae \in aH \implies a \in bK \implies a^{-1} \in bK$$

and $b = be \in bK$. By **Property 9 of Lemma 7.1**, bK is a subgroup of G because $b \in G$. Now see that

$$x = a^{-1}ax \in bK \implies bx \in bK \implies x \in K$$

as desired, and the first step above is by closure. Thus, $H \subseteq K$.

$\Leftarrow$ The proof in the other direction is very similar to show that $K \subseteq H$. Simply replace all a with b and vice versa and H with K and vice versa.

Thus, $H = K$, as desired. □

10. Let a and b be nonidentity elements of different orders in a group G of order 155. Prove that the only subgroup of G that contains a and b is G itself.

Proof. Let $a, b \in H$ where $a, b \neq e$, $|a| \neq |b|$, $H \subseteq G$, and $|G| = 155$. We want to show that H must be G . Note that the order of G is finite because $|G| = 155$ and $H \subseteq G$. Therefore, by **Lagrange's Theorem**, $|H| \mid |G|$. The only positive divisors of 155 are 1, 5, 31, 155. Therefore,

$$|H| \in \{1, 5, 31, 155\}$$

Note that $e, a, b \in H$ is given, so $|H| \neq 1$. Now suppose $|H| = 5$. Then by **Corollary 2** which states that in a finite group, the order of each element of the group divides the order of the group, this implies that $|a| \mid |H|$ and $|b| \mid |H|$, and therefore, $|a|, |b| \in \{1, 5\}$. However, $a \neq e$ and $b \neq e$ which implies that $|a| = 5$ and $|b| = 5$, which contradicts the given statement that a and b are of different orders. A similar argument can be applied to show that $|H| \neq 31$ because 31 is also prime. Therefore, $|H| = 155$, and since $H \subseteq G \implies H = G$ must be true, and so the only subgroup of G that contains a and b is G itself, as desired. $\square$

- 32.** Suppose that G is a group of order $2p$ where p is an odd prime. What are the possibilities for the number of elements of order 2?

Proof. By **Theorem 7.3**, G is isomorphic to $\mathbb{Z}_{2p}$ or D_p . Let's proceed by taking cases. Note that isomorphisms preserve order.

- (i) Suppose $G \approx \mathbb{Z}_{2p}$. Note that the only element of order 2 in $\mathbb{Z}_{2p}$ is p , and so the number of elements of order 2 are 1.
- (ii) Suppose $G \approx D_p$. Note that D_p consists of p reflections and p rotations. The only rotation of order 2 is 180 degrees, and all reflections have order 2. Thus, the number of elements of order 2 are $p + 1$.

□

42. Prove that a group of order 63 must have an element of order 3.

Proof. Let G be a group of order 63, which is finite. We want to show that there exists some element $a \in G$ such that $|a| = 3$, so on the contrary, assume that there does not exist an element of order 3. By **Corollary 2**, $|a|/63$. Therefore,

$$|a| \in \{1, 3, 7, 9, 21, 63\}$$

Note that $|a| = 1 \implies a = e$, which doesn't have order 3. We assume that $\nexists a \in G$ s.t. $|a| = 3$. Further observe that $|a| \neq 9$ because then the element $|a^3| = 3$, which contradicts our given assertion. The same argument can be applied to 21 and 63 because then $|a^7| = 3$ and $|a^{21}| = 3$. Therefore, the order of a must be 7 for all elements in $G \setminus \{e\}$. Note that $|G \setminus \{e\}| = 62$. By **Corollary 4.1**, the number of elements of order 7 in $G \setminus \{e\} \implies |G \setminus \{e\}| = 62$ must be a multiple of $\phi(7) = 6$. However, 62 is not a multiple of 6, and so we have arrived at a contradiction. Therefore, there exists an element a of order 3 in G . $\square$

62. Prove that A_5 has no subgroup of order 15 or 20.

Proof. Note that A_5 is the alternating group of even permutations on 5 elements. For each permutation to be even, there must be an even number of 2-cycles. Therefore, the elements of A_5 can be

- The identity $\implies 1$ of these
- 3-cycles $\implies \binom{5}{3} * 2 = 20$ of these
- 5-cycles $\implies \frac{5!}{5} = 24$ of these
- Disjoint 2-cycles $\implies \frac{5!}{2^2 * 2! * 1} = 15$ of these

and we confirm for our sanity that this does sum to 60, the total number of elements in A_5 . Now, on the contrary, suppose that there is a subgroup H of order 20 in A_5 . Consider $a \notin H, |a| = 5$. Our aim is to show that every element of order 5 is in H . As there are 24 5-cycles, this will prove that our assumption is wrong in this case. We can decompose A_5 into the disjoint union of cosets

$$A_5 = H \cup aH \cup a^2H$$

Now consider the element a^3 , which must be in A_5 , and therefore must be in either H, aH , or a^2H .

- Suppose $a^3 \in H \implies (a^3)^2 \in H \implies a^6 = a \in H$, which is a contradiction
- Suppose $a^3 \in aH \implies a^2 \in H \implies (a^2)^3 = a^6 = a \in H$, which is a contradiction
- Suppose $a^3 \in a^2H \implies a^{-2}a^3 \in a^{-2}a^2 \in H \implies a \in H$, which is a contradiction.

Thus, $a^3 \notin A_5$, which itself is a contradiction, and this means $a \in H$ for all 5-cycles a . Therefore, considering only 5-cycles, $|H| \geq 24$ (not including the identity) and so H cannot be of order 15 or 20, as desired. $\square$